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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/163,169	02/01/2023	Hayley WARSINSKE	49407-752.301	5871

160601 7590 02/29/2024

Freenome / WSGR
650 PAGE MILL ROAD
PALO ALTO, CA 94304-1050

EXAMINER
NEGIN, RUSSELL SCOTT

ART UNIT	PAPER NUMBER
1672	

NOTIFICATION DATE	DELIVERY MODE
02/29/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentdocket@wsgr.com

Office Action Summary	Application No. 18/163,169	Applicant(s) WARSINSKE et al.	
	Examiner RUSSELL S NEGIN	Art Unit 1672	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 14 February 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-10,12,14,16-18 and 20-24 is/are pending in the application.

5a) Of the above claim(s) 3-6 and 8-10 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,7,12,14,16-18 and 20-24 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 14 February 2024 has been entered.

Comments

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Claims 3-6 and 8-10 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected species, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 15 May 2023.

Claims 1, 3-10, 12, 14, 16-18, and 20-24 are pending in the application.

Claims 1, 7, 12, 14, 16-18, and 20-24 are examined in the instant Office action.

Information Disclosure Statement

The IDS filed on 2/16/2024 has been considered.

Withdrawn Rejections

The rejections under 35 U.S.C. 103 are withdrawn in view of amendments filed to the instant set of claims on 14 February 2024.

Claim Rejections - 35 USC § 112(b) - Indefiniteness

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

The following rejections are newly applied:

Claims 1, 7, 12, 14, 16-18, and 20-24 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Each of the claims 1 and 20 recite a second set of samples obtained or derived from subjected without advanced adenoma. It is unclear as to whether a patient without advanced adenoma is a control (i.e. healthy) patient, or a patient with a malignant or benign tumor. For the purpose of examination, it is interpreted that a patient without

advanced adenoma can be either a control (i.e. healthy) patient, or a patient with a malignant or benign tumor.

Each of the claims 1 and 20 recite a fourth set of samples obtained or derived from subjected without benign polyp. It is unclear as to whether a patient without benign polyps is a control (i.e. healthy) patient without any polyps, or a patient with a malignant tumor. For the purpose of examination, it is interpreted that a patient without advanced adenoma can be either a control (i.e. healthy) patient without polyps, or a patient with a malignant tumor.

Claim 24 recites a sixth set of samples obtained or derived from subjected without colorectal cancer. It is unclear as to whether a patient without advanced adenoma is a control (i.e. healthy) patient, or patient with a different type of cancer. For the purpose of examination, it is interpreted that a patient without colorectal cancer can be either a control (i.e. healthy) patient, or a patient with a different type of cancer.

In view of these indefiniteness rejections, it is interpreted that there is overlap between the six sets of biological samples. For example, a healthy patient without any lesions can have samples that satisfy the second, fourth, and sixth sets of samples.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

The following rejection is necessitated by amendment:

Claim(s) 1, 7, 12, 14, and 16-24 is/are rejected under 35 U.S.C. 103 as being unpatentable over Vogelstein et al. [WO 2019/067092 A1] in view of Angelino et al. [Annals of Oncology, volume 30, supplement 5, October 2019, abstract 146P] in view of Janjua et al. [APMIS, volume 123, 2015, pages 298-304].

Claim 1 is drawn to a method of detecting colorectal cancer in a subject using a computer specifically programmed to detect cancer. The method comprises obtaining a protein profile of at least three proteins from a given list. The method comprises detecting an increased risk of the colorectal cancer in the subject. The detecting comprises processing the protein profile using a trained machine learning model. The method requires that the trained machine learning model is trained with training data comprising a first set of biological samples obtained or derived from subjects with advanced adenoma and a second set of biological samples obtained or derived from subjects without advanced adenoma. The data comprises a third set of data obtained

or derived from subjects with a benign polyp. The data comprises a third set of data obtained or derived from subjects without a benign polyp. The result is an output value indicative of the increased risk of colorectal cancer. The method requires performing a colonoscopy when there is an increased risk of colorectal cancer.

Claim 7 limits the at least six proteins to comprise FLT3L.

Claim 14 teaches obtaining a blood sample from the subject.

Claim 18 recites treatment using surgery or chemotherapy.

Claim 20 is drawn to similar subject matter as claim 1.

Claim 19 is drawn to similar subject matter as claim 1, except claim 19 is drawn to a computer.

Claim 23 is further limiting comprising a combination of any six proteins from a given list.

The document of Vogelstein et al. studies materials and methods for assessing and treating cancer [title]. Paragraph 181 of Vogelstein et al. teaches that a type of cancer studied is colorectal cancer. Paragraph 453 of Vogelstein et al. teaches the computer limitations of the claims. Paragraph 193 of Vogelstein et al. teaches that EGF, FGF-2, and FLT3L are biomarkers of interest. Paragraph 198 of Vogelstein et al. teaches that IL-6 and IL-8 are biomarkers of interest. Paragraph 215 of Vogelstein et al. teaches that IL-2 is a biomarker of interest. Paragraph 185 of Vogelstein et al. teaches that MIP-3a is a biomarker of interest. Paragraph 286 of Vogelstein et al. teaches obtaining a blood sample from the subject. Paragraphs 43, 687, and Figure 8 of Vogelstein et al. teach performing machine learning based on a trained model to differentiate subjects with and without colorectal cancer. Paragraph 859 of Vogelstein

et al. teaches treating the subject with surgery or chemotherapy. Paragraph 160 of Vogelstein et al. teaches performing colonoscopies in patients with increased risk of colorectal cancer.

Vogelstein et al. does not stratify the machine learning based on the presence or absence of advanced adenomas or benign polyps.

The abstract of Angelino et al. studies discovery of an immunotranscriptomics signature in blood for early colorectal cancer detection [title]. The Methods and Results paragraph of Angelino et al. teaches testing blood to find biomarkers relevant for machine learning used to differentiate risk of colorectal cancer in patients with colorectal cancer, patients with advanced adenomas, and patients without both colorectal cancer and advanced adenomas. The Methods and Results paragraph of Angelino et al. teaches 218 control patients without any lesions.

Vogelstein et al. and Angelino et al. do not stratify the machine learning based on the presence or absence of benign polyps.

The document of Janjua et al. studies hyperplastic polyps of the colon and rectum with reclassification, BRAF and KRAS status in index polyps and subsequent colorectal carcinoma [title]. The abstract, introduction, and Tables of Janjua et al. teach patients with benign polyps.

With regard to claims 12 and 17, paragraph 1132 of Vogelstein et al. differentiates/stratifies the types of colorectal cancer wherein one of the subtypes comprises colorectal adenoma.

With regard to claim 16, paragraph 71 of Vogelstein et al. teaches that the trained machine learning model comprises SVM.

With regard to claims 21-22, paragraph 286 of Vogelstein et al. teaches obtaining a blood sample from the subject for further biological assays. Paragraph 37 of Vogelstein et al. teaches using blood plasma.

With regard to claim 24, the Methods and Results paragraph of Angelino et al. teaches testing blood to find biomarkers relevant for machine learning used to differentiate risk of colorectal cancer in patients with colorectal cancer, patients with advanced adenomas, and patients without both colorectal cancer and advanced adenomas.

It would have been obvious to someone of ordinary skill in the art at the time of the effective filing date of the instant application to modify the machine learning that differentiates between control and colorectal cancer patients of Vogelstein et al. by further including patients with advanced adenomas as a category of stratification as in Angelino et al. wherein the motivation would have been that analyzing patients without colorectal cancer but with advanced adenomas provides an earlier warning sign of increased risk of colorectal cancer [title and abstract of Angelino et al.]. There would have been a reasonable expectation of success in combining Vogelstein et al. and

Angelino et al. because both studies are analogously applicable to the field of understating colorectal cancer using machine learning.

It would have been obvious to someone of ordinary skill in the art at the time of the effective filing date of the instant application to modify the machine learning that differentiates between control and colorectal cancer patients of Vogelstein et al. and further including patients with advanced adenomas as a category of stratification as in Angelino et al. by use of the benign polyps of Janjua et al. wherein the motivation would have been that the presence of benign polyps is an additional form a stratification useful in predicting the onset colorectal cancer [abstract and introduction of Janjua et al.]. There would have been a reasonable expectation of success in applying the stratification of Vogelstein et al. and Angelino et al. to the benign polyps of Janjua et al. because the stratification analysis of Vogelstein et al. and Angelino et al. is general and robustly applicable to other lesions related to colorectal cancer (i.e. benign polyps in Janjua et al.).

Response to arguments:

Applicant argues that the amendments to the claims overcome the prior art rejections. In response, the prior art of Janjua et al. has been added to make obvious the amended limitations of the claims.

Related Prior Art

The document of Singh et al. [Cancer Cell, volume 1, 2002, pages 203-209; on IDS] teaches using specific genes as biomarkers and machine learning to detect

prostate cancer. However, Singh et al. does not teach the recited protein panel and pertains to prostate cancer (and not colorectal cancer).

The document of Golub et al. [Science, volume 286, 1999, pages 531-537; on IDS] teaches molecular classification of cancer using class discovery and class prediction by gene expression monitoring. However, Golub et al. pertains to leukemia (and not colorectal cancer) and does not teach the recited protein panel.

E-mail Communications Authorization

Per updated USPTO Internet usage policies, Applicant and/or applicant's representative is encouraged to authorize the USPTO examiner to discuss any subject matter concerning the above application via Internet e-mail communications. See MPEP 502.03. To approve such communications, Applicant must provide written authorization for e-mail communication by submitting the following statement via EFS-Web (using PTO/SB/439) or Central Fax (571-273-8300):

> Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned and practitioners in accordance with 37 CFR 1.33 and 37 CFR 1.34 concerning any subject matter of this application by video conferencing, instant messaging, or electronic mail. I understand that a copy of these communications will be made of record in the application file.

Written authorizations submitted to the Examiner via e-mail are **NOT** proper. Written authorizations must be submitted **via EFS-Web** (using PTO/SB/439) **or Central Fax** (571-273-8300). A paper copy of e-mail correspondence will be placed in the patent application when appropriate. E-mails from the USPTO are for the sole use of the intended recipient, and may contain information subject to the confidentiality requirement set forth in 35 USC § 122. See also MPEP 502.03.

Conclusion

No claim is allowed.

Any inquiry concerning this communication or earlier communications from the Examiner should be directed to Russell Negin, whose telephone number is (571) 272-1083. This Examiner can normally be reached from Monday through Thursday from 8 am to 3 pm and variable hours on Fridays.

If attempts to reach the Examiner by telephone are unsuccessful, the Examiner's Supervisor, Larry Riggs, Supervisory Patent Examiner, can be reached at (571) 270-3062.

/RUSSELL S NEGIN/
Primary Examiner, Art Unit 1672
25 February 2024